AI and Society · SC3098 · Group Design Project

Introduction to Robo-Advisory

Joel Quek · Lecturer, CCDS

College of Computing and Data Science · Nanyang Technological University

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Agenda

How does software go from “I have $50,000” to a responsible portfolio recommendation?

1

What can money actually be invested in?

Stocks, bonds, cash-like assets, funds and ETFs.

2

How do those become a portfolio?

Allocation, diversification, core versus satellite.

3

Why should two people get different portfolios?

Horizon, liquidity, willingness and capacity for loss.

4

Where should a AI chatbot powered robo-advisor fit, and where should it not?

Conversation and explanation versus deterministic rules.

Tomorrow's lab, I will cover working with financial data, returns, drift and rebalancing.

Part 1

What are we actually investing in?

Before the robo-advisor can advise, words like stock, bond, ETF and portfolio need to mean something.

You have $50,000. What can it actually become?

At the most basic level, you can own, lend, or keep money liquid

OWN

Stock / equity

Buy a claim on a company. Your value rises or falls with the business and the market.

LEND

Bond / fixed income

Lend money to a government or company. You receive promised payments, but still take credit and interest-rate risk.

PARK

Cash / T-bill

Keep money highly liquid and relatively stable. You give up some growth potential for near-term safety.

Most portfolios are combinations of these basic economic claims, not seven unrelated tickers.

Wait: when I buy a stock, what did I actually buy?

A stock is ownership, not a loan



Imagine the company is divided into many tiny ownership claims.

You own a slice

A share gives you a residual claim on the company. If the business becomes more valuable, your share may become more valuable too.

You are not promised repayment

Unlike a bond, the company does not owe you your purchase price back on a fixed date.

In this project

We do not pick individual companies. SPY gives broad equity exposure through an ETF.

So how does a stock make, or lose, money?

There are two ordinary routes, and neither is guaranteed

1 · The price changes

Investors continually revise what they think the company is worth. Better expected profits can push the price up; bad news can push it down. The market can also move even when the company itself has not changed much.

2 · The company may pay dividends

Some companies distribute part of their profits to shareholders. Others reinvest instead. A stock can therefore deliver cash income, price gains, both, or neither.

The important bit for robo-advisory:
equity offers meaningful long-run growth potential, but the client must be able and willing to live through large interim falls.

A bond is different: you are lending, not owning

The borrower promises payments; you still take risk



The basic contract

You lend money to a government or company. In return, the bond promises interest payments and repayment of principal at maturity.

Why can a bond still lose money?

- Interest rates rise → existing bond prices can fall.
- The borrower could become less creditworthy.
- A bond ETF holds many bonds, so its market value moves every day.

Less volatile than equities does not mean “cannot fall”.

And “cash” is not just money under a mattress

In this simulation, BIL is our cash-like sleeve

Cash-like goal

Preserve liquidity and avoid large short-term swings.

BIL

An ETF holding very short-term U.S. Treasury bills.

What you gain

High liquidity and very small price swings compared with stocks.

What you give up

Lower growth potential; returns can lag inflation and future rates can change.

In the project, “cash” is shorthand for a cash-like market instrument, not literal banknotes.

Three building blocks, three different jobs

The project's core portfolios mix SPY, AGG and BIL

SPY

Broad equity ETF

Role: long-term growth

Main risk: market drawdowns

AGG

Broad bond ETF

Role: income + stabiliser

Main risk: rate / credit moves

BIL

T-bill ETF

Role: liquidity + safety

Main risk: inflation / reinvestment

None of these is “the best asset”. They are useful for different reasons.

Okay... then what is an ETF?

An ETF is a container, not a risk level

Exchange-traded fund

A fund pools many underlying investments. One ETF unit gives you a tiny slice of that pool, and the unit itself trades on an exchange during the day.

Broad ETF

SPY spreads exposure across hundreds of large U.S. companies.

Concentrated ETF

SMH still is an ETF, but it concentrates exposure in semiconductors.

ETF tells you the wrapper. You still need to ask what is inside.



S&P 500 ≠ SPY

Index, fund and ETF describe different things

INDEX

A rulebook / benchmark

S&P 500 specifies which large U.S. companies are in the index and how they are weighted.

FUND

A pool of assets

A fund actually holds securities in an attempt to follow an investment objective.

ETF

How the fund is packaged

An ETF is a fund whose units can be bought and sold on an exchange. SPY is an ETF designed to track the S&P 500.

This distinction matters because the robo-advisor recommends investable products, not abstract indices.

Now we can finally define a portfolio

A portfolio is a collection of investments and the weights assigned to them

Example: $50,000 in the project's Balanced target



The weights are the key idea.

The lab will later turn these dollar targets into units, revalue them as prices move, and show how the weights drift away from target.

Why not just buy the thing we think will do best?

Because concentration makes one mistake, or one bad event, matter too much

Company-specific shock



One firm has a scandal, product failure, or lawsuit. In a diversified basket, that one firm is only a small piece.

Market-wide shock



A recession or market panic can hit many holdings together. Diversification reduces single-name risk; it does not eliminate market risk.

Diversification is not “something always goes up”.
It is “one thing going wrong should not decide everything”.

Risk and return: what does “risk” actually mean here?

Higher expected return normally requires accepting more uncertainty, not a guaranteed payoff



For this module, “risk” is broader than volatility.

- **Volatility**
  How much market value moves around.
- **Drawdown**
  How far the value can fall from a previous high.
- **Suitability**
  Whether the client can tolerate the loss and wait long enough for uncertainty to resolve.

A single stock can add concentration risk without promising a higher expected return.

Core and satellite: broad first, concentrated second

The satellite is optional exposure, not a replacement for diversification

CORE · the bulk of the portfolio

Broad holdings that do most of the work. In this project: SPY + AGG + BIL. The point is breadth, stability of design, and a clear baseline allocation.

SATELLITE · small, capped, optional

Concentrated sector or thematic exposure, taken in small amounts on top of the core. These are also ETFs, but each tracks a single industry, so they swing harder than the broad market.

The four satellite ETFs

XLK	Technology
XLV	Healthcare
XLE	Energy
SMH	Semiconductors

Why cap it?

A sector ETF may outperform or underperform. The cap exists because concentration risk is real, and access is gated behind a knowledge check.

Part 2

Who should hold what?

Robo-advisory begins when the investment question becomes a person question.

DISCUSSION

Same age, same salary, different advice?

Two 40-year-olds both say they want “good returns”. One is advised a high-equity portfolio; the other a much more cautious one. What could justify that?

- **Horizon**
- **Liquidity**
- **Savings / commitments**
- **Comfort with loss**

The answer is not “one of them likes better returns”. The difference is what loss they can live with, financially and emotionally.

Time horizon changes what losses you can wait through

The issue is not that a market fall becomes magically permanent; it is whether you are forced to sell before recovery

Need the money in ~2 years



A large drawdown just before the goal can force you to realise the loss.

Need the money in ~20 years



More time means more opportunity to wait through bad periods, although recovery is never guaranteed.

Same person, same wealth, different goal date
→ **potentially different suitable portfolio.**

Liquidity changes what losses you can afford to wait through

Money you may need soon should not be treated like money you can leave alone for years

Money I may need this year

Emergency spending, tuition, a house deposit, near-term commitments. If markets fall, this money may need to be withdrawn anyway, so the room for market risk is small.

Money I can leave for the long term

A long-dated goal can usually tolerate more interim fluctuation because the money is not needed immediately.

Why the robo-advisor asks

A client who says “I can handle losses” but may need half the money within 12 months still has a liquidity constraint.

Willing to lose ≠ able to lose

Risk attitude and financial capacity answer different questions

Risk appetite · "How would I feel?"

A preference. Some people cannot sleep through a 10% drop; others are comfortable seeing a much larger temporary loss.

Risk capacity · "Can my finances absorb it?"

A constraint. Income, savings, commitments and dependants determine how damaging a loss could actually be.

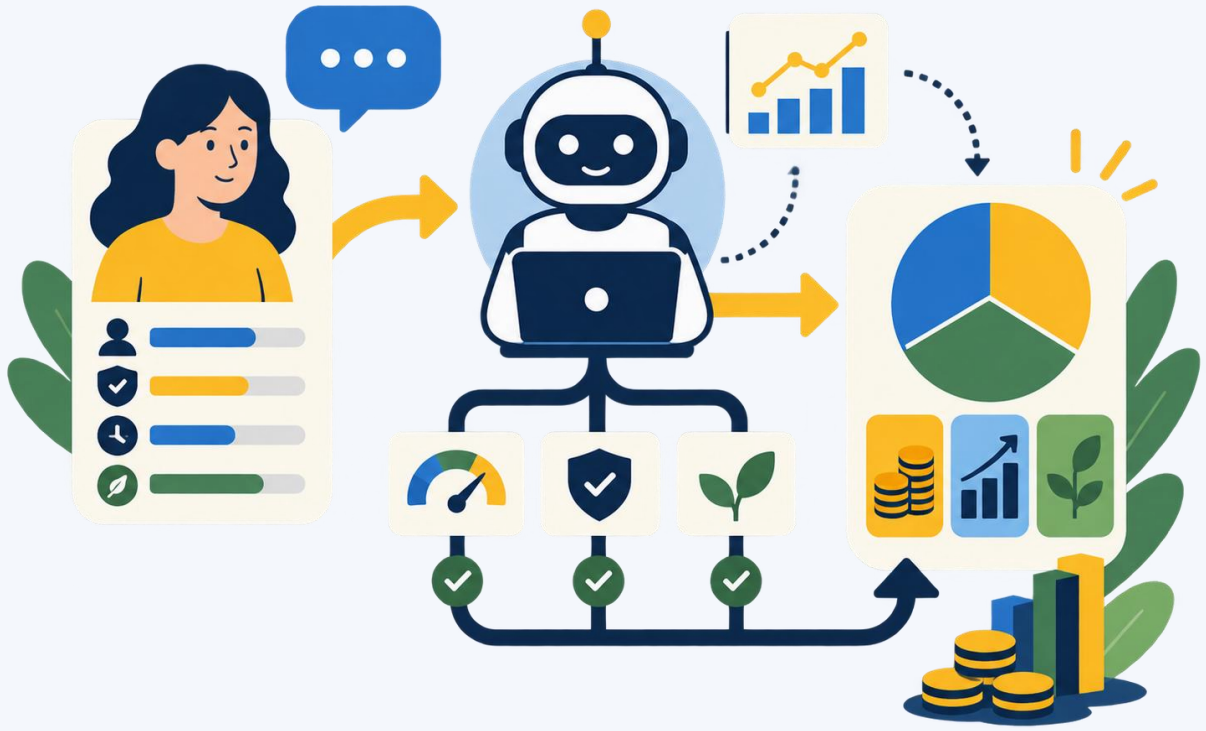
Preference never overrides capacity.

A responsible recommendation respects both, which is why the project captures both kinds of information.

So what is the robo-advisor actually doing?

Not predicting the market, but matching a person to an allowed portfolio

Ask	Who are you?
Assess	What risk fits?
Match	What is allowed?
Invest	Put money to work
Watch	Revalue and adjust



The “robo” part is the automation of the advisory workflow, not a robot that knows the future.

Which parts should the LLM actually decide?

This is the Responsible AI boundary that matters in your build

LLM / conversational layer

- ✓ Converse naturally
- ✓ Clarify incomplete or inconsistent answers
- ✓ Retrieve policy / product information
- ✓ Explain decisions in plain language
- ✓ Refuse and redirect when needed

Deterministic platform logic

- ✓ Compute the risk score
- ✓ Apply suitability caps and gates
- ✓ Decide eligible portfolios
- ✓ Execute and value holdings
- ✓ Compute rebalancing trades
- ✓ Record the audit trail

The LLM can explain an entitlement. It should not invent one.

The same boundary keeps the secrets:
the client sees the risk category, never the numeric score; score and formula live in the backend, and extracting either through the chat is a scored red-team attack.

One point to consider:

“I know the risks. Just give me the highest-return portfolio.”

The client is articulate, confident and persistent. Should the chatbot obey because the user explicitly asked for it?

Correct direction:

the system may explain and offer a more cautious choice, but it should not bypass the eligibility rules because the user is persuasive.

Refusing well is part of the advisory job.

From person to portfolio: the end-to-end project flow

The appendices define behaviour; the application connects the pieces

A	**Profile**	Client form and conversation capture the relevant facts.
B	**Score**	Backend computes the reproducible risk category. The numeric score and the formula are confidential: the client is told the category only.
E	**Suitability**	Policy caps, gates and eligibility rules decide what may be recommended. A client who needs the money soon, or cannot tolerate loss, is capped below their score, or advised not to invest at all.
G	**Explain**	Recommendation comes with grounded explanation and the mandatory disclosure block.
C	**Invest**	Simulated $50,000 becomes holdings at the chosen session date.

Audit trail runs through the whole flow → inputs, rules, retrieved evidence, outputs, refusals and transactions.

Why do all these rules exist?

Because financial advice can harm people when the system is confident, wrong, or too helpful

1

Know the client

Collect enough information before advising.

2

Suitability

Recommend only what fits the client and the policy.

3

Disclosure

Explain key risks, assumptions and limitations.

4

Records

Keep evidence of what the system knew and why it acted.

For AI & Society, the interesting question is not merely “does the chatbot answer?”

It is whether the whole system stays bounded, transparent, consistent and auditable when the user pushes it.

THE ONE IDEA TO KEEP

A robo-advisor does not predict the market.

It matches a person to a bounded portfolio, within rules, and can manage the portfolio automatically.